PhySci/MiMG/CaSB M178

Homework 2

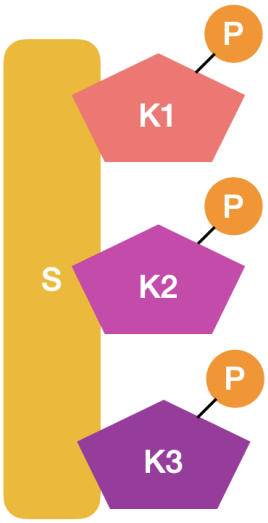
Due: 10/13/26 at 2:00PM PDT

Notes: This homework involves performing simulations of the Scaffold Proteins we've been discussing in the last two class meetings. On the same assignment on Bruinlearn where you obtained this document, you will also find a file called "HW2_template.ipynb" that contains a template Jupyter notebook that you can use as a starting point to complete the questions below. Please modify this notebook and use it as the starting point for answering the following problems.

To submit your homework, please answer the questions below. Note that you will have to **paste in several graphs** that you generate using the Jupyter notebook. After completing the questions, **save this document as a PDF** and upload it to Gradescope. **You must also upload the Jupyter notebook to Bruinlearn**. Finally, do not forget to **MARK YOUR QUESTIONS** when uploading to Gradescope.

Part 1: Intro to Scaffold Models

In class, we talked about scaffolding proteins. Consider the following scaffold protein example that accommodates three different kinase species:



Like in last week's homework, upon a receptor binding event at the cell surface, the first kinase species, K_1, is changed (phosphorylated) to its active form, K_1p (this activation process is not explicitly modeled here). This active kinase species can bind the scaffold protein, S, and then can phosphorylate the next kinase species, K_2, if it is present on the scaffold protein as well. K_2p can then

activate K_3 if both are bound to the scaffold protein. We will assume all kinase species (whether bound to the scaffold or not) undergo a first-order dephosphorylation at the same rate.

We have the following monomer species in our model: S, K_1p, K_2, K_2p, K_3, K_3p.

1. (3 points) Write down the number of dimers, trimers, and tetramers in this model given unordered, independent binding.

K1p has one modeled form, K2 can be K2 or K2p, K3 can be K3 or K3p.
The five dimers are SK1p, SK2, SK2p, SK3, SK3p
Dimer: Scaffold + one kinase, 1 + 2 + 2 = 5
Trimers: Scaffold + two kinase types, 1 x 2 + 1 x 2 + 2 x 2 = 8
Tetramers: Scaffold + all three kinase types, 1 x 2 x 2 = 4
Overall: There are 5 dimers, 8 trimers, and 4 tetramers. Each complex has one scaffold, and the inactive and phosphorylated forms of K2 and K3 count as distinct species.

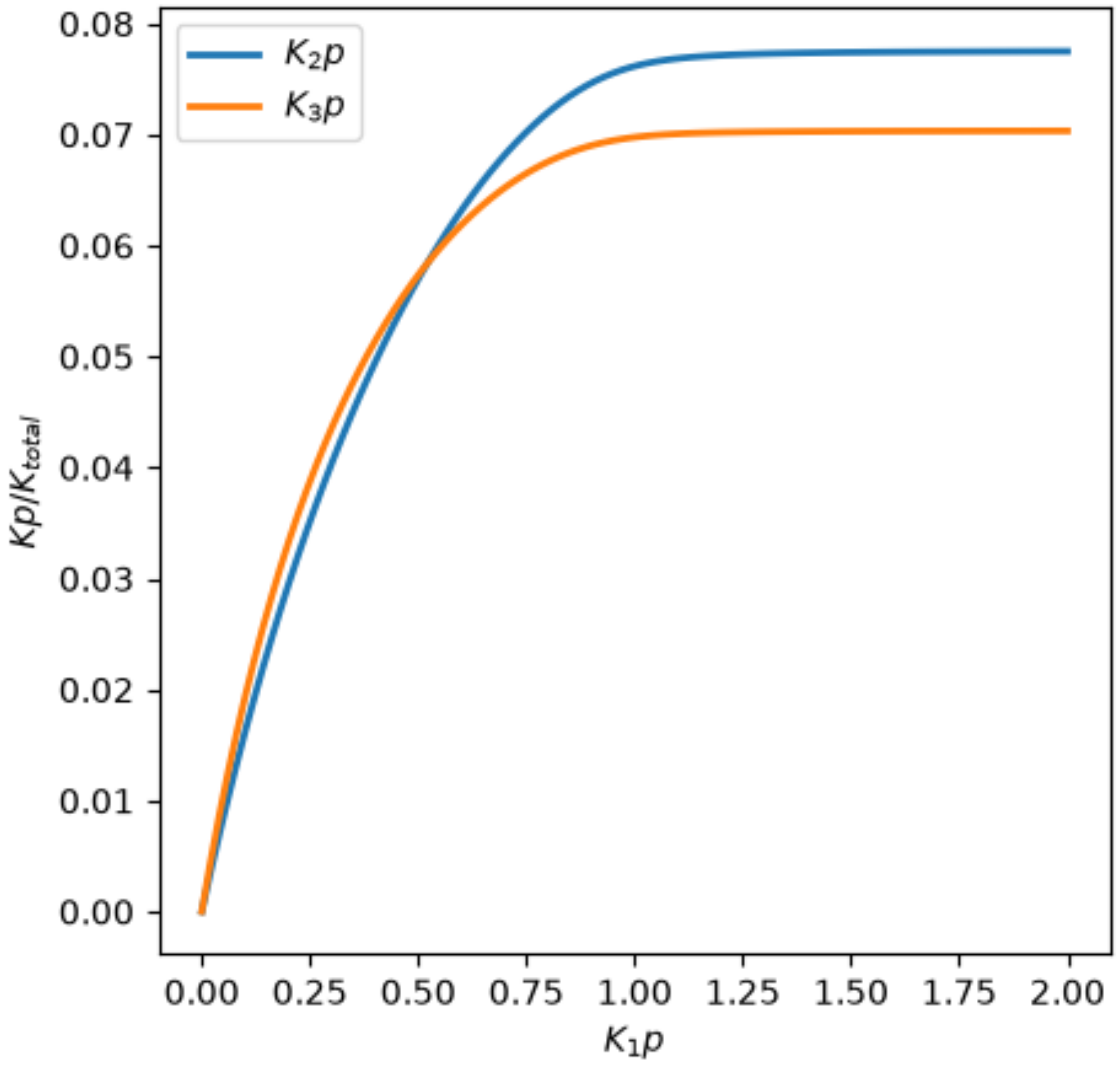
The unordered scaffold model with three kinases has been implemented for you in the section “scaffold model”. We will explore the amount of output, K_3p, present at steady state for varying values of input, K_1p, under the section “STEADY-STATE responses of scaffold model”.

Note that the parameters are set at the beginning of this section of code as well as the initial conditions. The rates of association and dissociation of the kinases on to and off the scaffold protein are all $10\mu M^{-1}s^{-1}$ and $0.1s^{-1}$, except the association of the third kinase species (in either its phosphorylated or dephosphorylated form) has a rate of $1\mu M^{-1}s^{-1}$ (k_a3). All of the downstream kinases are in the inactive form at the start and the scaffold protein concentration is equal to the kinase concentrations (1µM), except the third kinase species is ten times greater in concentration. The differences in rates for the third kinase species allow for more free K_3p to be present at steady state. The kinase must be free from the scaffold to act downstream in the nucleus.

This section of the code then changes the initial value of K_1p, from 0 to 2 (2 times its default value of 1). It then runs a simulation of the scaffold model with those parameters, and saves the result of the final time point. After doing this, it

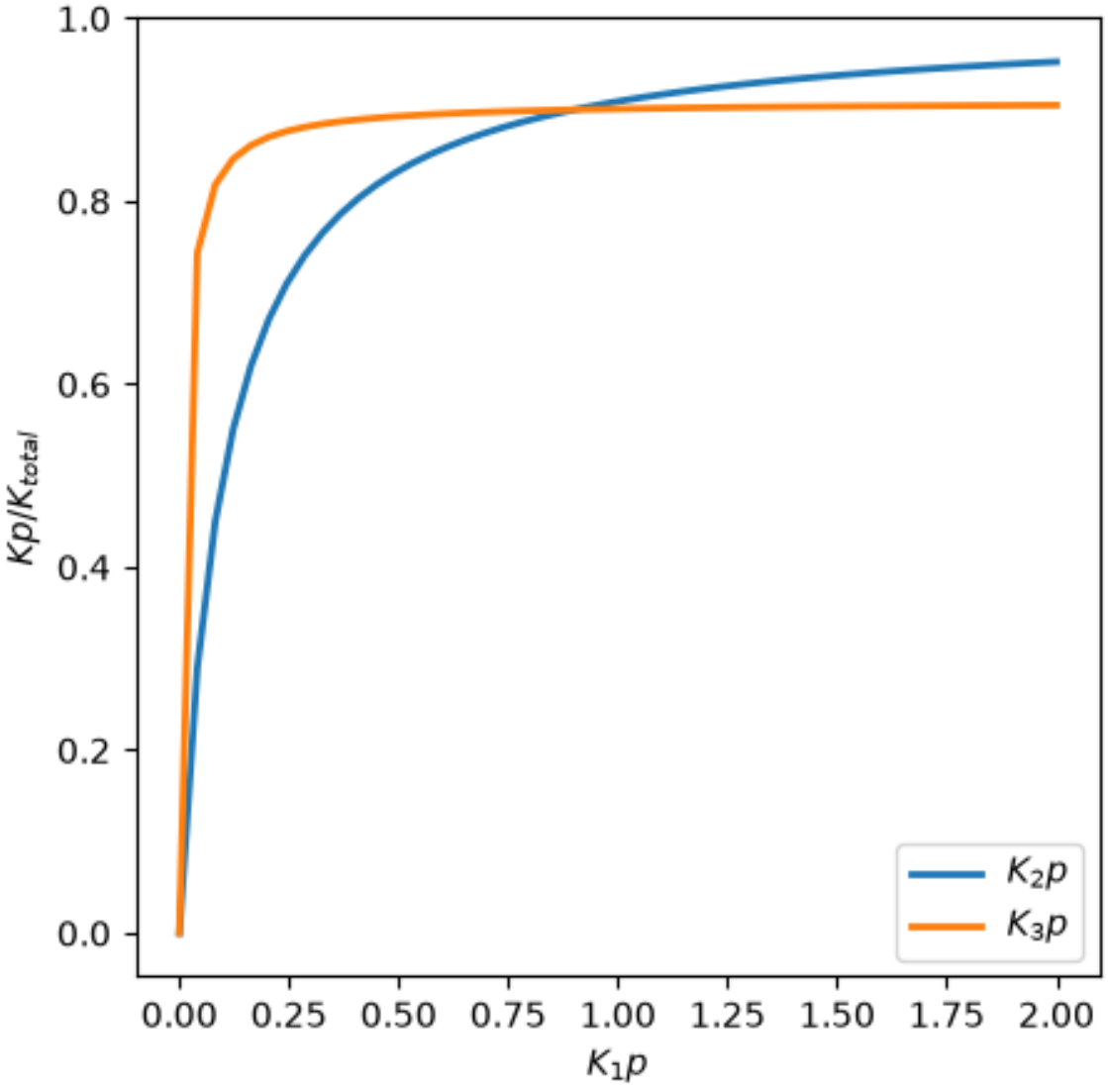
generates a plot of the steady-state value of $K_3p/K_{3\,total}$ as a function of the parameter, K_1p.

2. (3 points) Run the "STEADY-STATE responses of scaffold model" code and paste the resulting graph below.



Next under the section of code “no scaffold model” a kinase cascade with three kinase species has been implemented similar to the models in last week’s homework. Again, in the section “STEADY-STATE responses” we will explore how varying K_1p changes the steady state value of $K_3p/K_{3\,total}$.

3. (3 points) Run the “STEADY-STATE responses” code and paste the resulting graph below.



4. (5 points) How does the presence of the scaffold protein alter the amount of needed to achieve maximum activation of the final level kinase? How does the scaffold protein alter the "signal amplification" provided by the cascade (compare the curves from questions 2 and 3)?

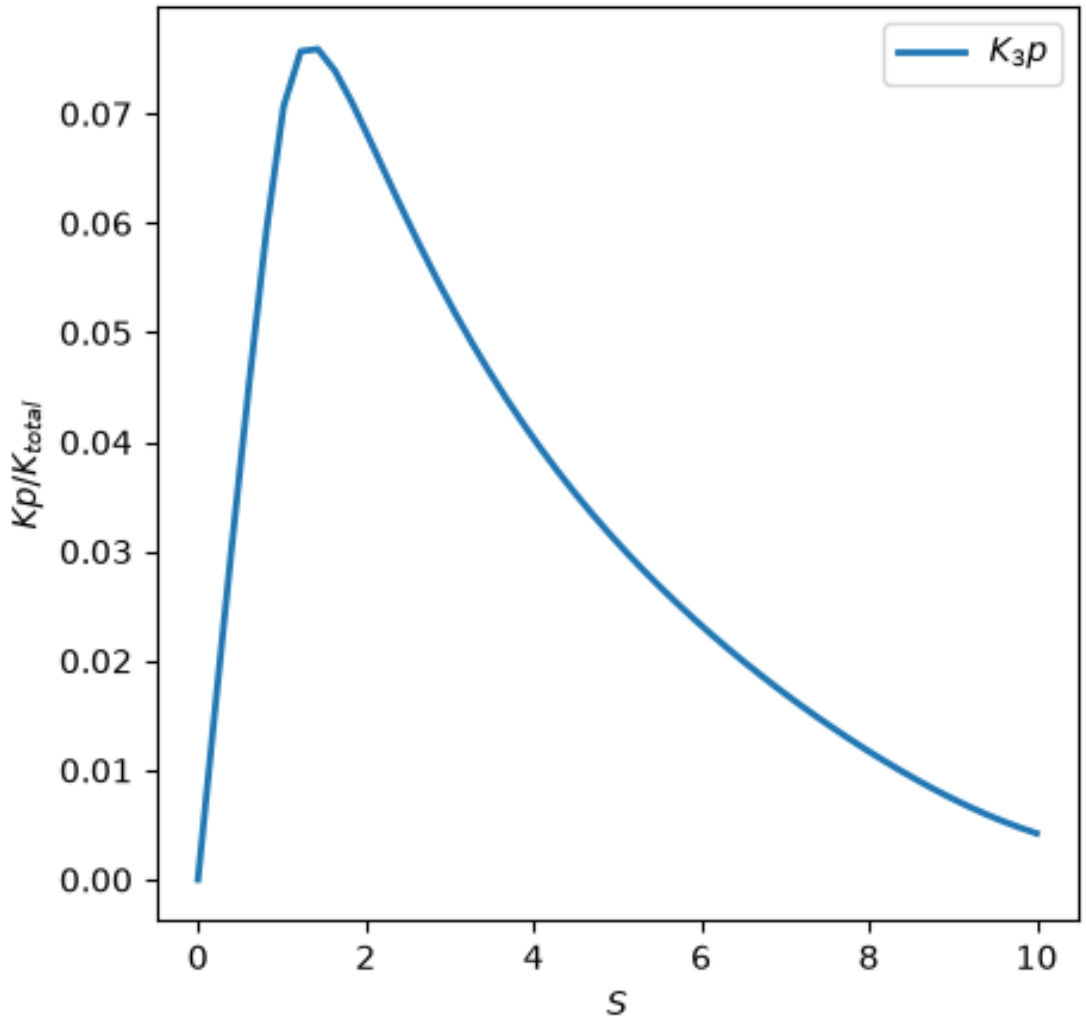
With the scaffold more K1p is needed to approach max activation of the final kinase. Without the scaffold, the K3p curve has a rapid rise with low input, and it reaches a much higher phosphorylated fraction.The scaffold is able to reduce signal amplification in this model because phosphorylation requires kinases to share a scaffold, and phosphorylated K3 can remain bound instead of contributing to the free output.

Part 2: Prozone Effect

Underneath the section where you plotted the "steady-state behavior," there is a section called "Prozone Effect." This is where we will examine the prozone effect or combinatorial inhibition discussed in class.

We will use the unordered scaffold model as defined previously. Notice again, that there is a section to define the parameters and initial conditions, which are the same as the previous simulations. Now there is a section that will vary the concentration of the scaffold protein, S, from 0 to 10 μM (ten times its default values of 1 μM) and then plot the steady state value of $K_3p/K_{3\,total}$ of the resulting simulations.

5. (3 points) Run the code, and paste the graph ("PLOT Steady State Responses") below.



6. (5 points) Explain in your own words what causes the initial rise in . Now explain what causes the decline in as S further increases.

Initially increasing scaffold concentration gives the kinases more places to bind together and phosphorylate one another. The fraction of free K3p hence increases. As scaffold concentration increases further, the limited kinases spread across different scaffold molecules, making productive kinase pairs actually less common. Phosphorylation then decreases, causing the decline in K3p/K3. This effect is the prozone effect which is also called combinatorial inhibition.

7. (5 points) What value of scaffold protein gives a maximal response? Predict how might this value might change if the kinase concentrations were smaller/larger. You can check your prediction by altering the initial conditions (K1p, K2).

We get a maximal response at a scaffold concentration of 1.43 μM. If kinase concentrations were smaller, we could expect that the optimal scaffold concentration would decrease, and if kinase concentration were larger we would expect an increase in optimal scaffold concentration. With fewer kinases, excess scaffold separates them across different molecules sooner, while more kinases allow a larger scaffold concentration before the separation reduces signaling.

Part 3: Ordered Binding Model

Thus far we have considered an unordered binding model. We will now consider an ordered binding model in the section “Ordered Binding”. In this model we still have a scaffold protein, S, and three kinase species. However, binding now occurs with the following rules:

1. Only K_1p can bind to S
2. Only K_2 or K_2p can bind to SK_1p
3. K_3 and K_3p can only bind SK_1pK_2p (at a reduced rate like the previous model)
4. Only K_3 and K_3p can unbind $SK_1pK_2pK_3$ and $SK_1pK_2pK_3p$ respectively
5. Only K_2 and K_2p can unbind to SK_1pK_2 and SK_1pK_2p respectively
6. K_1p can unbind from SK_1p

7. K_1p can phosphorylate K_2 if both species are bound to the scaffold (SK_1pK_2)
8. K_2p can phosphorylate K_3 if both species are bound to the scaffold ($SK_1pK_2pK_3$)
9. K_2p and K_3p can both dephosphorylate when unbound from the scaffold. They can also dephosphorylate when bound to the scaffold, except K_2p if K_3 or K_3p is also bound to the scaffold.
10. All phosphorylation and dephosphorylation reactions are first-order with rates k_p and k_u respectively
11. Association/binding events all occur with rate k_a and dissociation/unbinding events all occur with rate k_d, except K_3 and K_3p associate with the scaffold with reduced rate k_a3

In summary these are then the possible species in the model:
$S,\ K_1p,\ K_2p,\ K_3,\ K_3p,\ SK_1p,\ SK_1pK_2,\ SK_1pK_2p,\ SK_1pK2_pK_3,\ SK_1pK2_pK_3p$.

(See the question on the next page!)

8. (5 points) Write the change equations for all species in the model and add them to the code section "Ordered Binding" where it says fill in your change equations. Take a screenshot of your change equations and paste them below.

```
S_prime = -k_a*S*K1p + k_d*SK1p

K1p_prime = -k_a*S*K1p + k_d*SK1p

K2_prime = -k_a*SK1p*K2 + k_d*SK1pK2 + k_u*K2p

K2p_prime = -k_a*SK1p*K2p + k_d*SK1pK2p - k_u*K2p

K3_prime = -k_a3*SK1pK2p*K3 + k_d*SK1pK2pK3 + k_u*K3p

K3p_prime = -k_a3*SK1pK2p*K3p + k_d*SK1pK2pK3p - k_u*K3p

SK1p_prime = (
    k_a*S*K1p - k_d*SK1p
    - k_a*SK1p*K2 - k_a*SK1p*K2p
    + k_d*SK1pK2 + k_d*SK1pK2p
)

SK1pK2_prime = (
    k_a*SK1p*K2 - k_d*SK1pK2
    - k_p*SK1pK2 + k_u*SK1pK2p
)

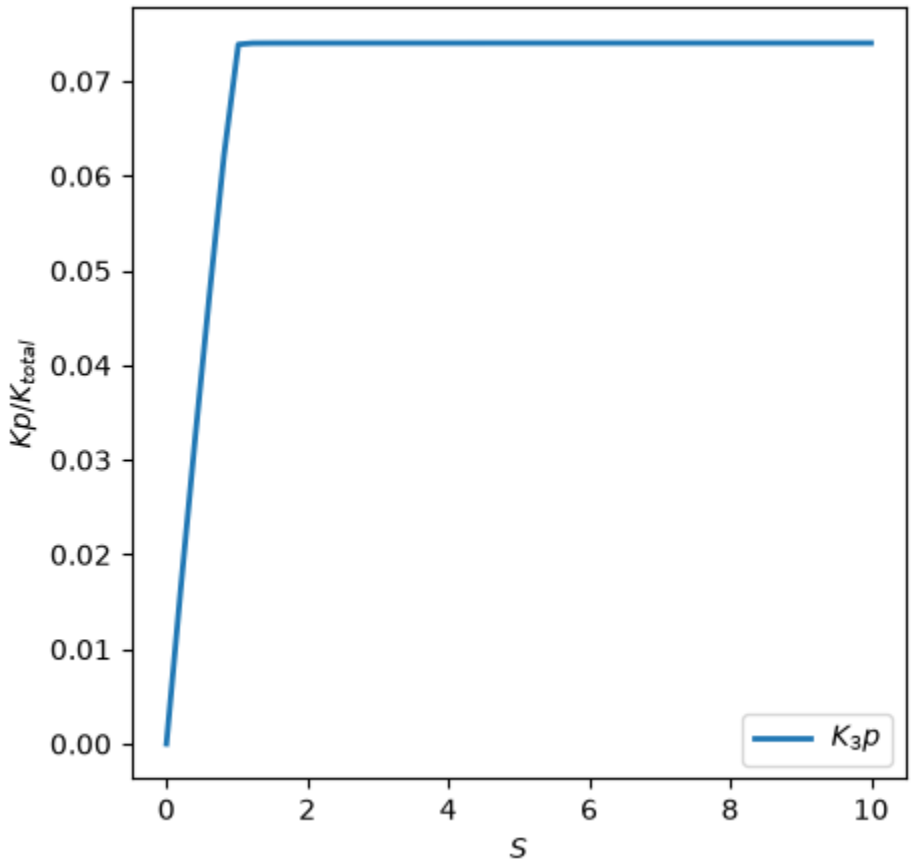
SK1pK2p_prime = (
    k_a*SK1p*K2p - k_d*SK1pK2p
    + k_p*SK1pK2 - k_u*SK1pK2p
    - k_a3*SK1pK2p*K3 - k_a3*SK1pK2p*K3p
    + k_d*SK1pK2pK3 + k_d*SK1pK2pK3p
)

SK1pK2pK3_prime = (
    k_a3*SK1pK2p*K3 - k_d*SK1pK2pK3
    - k_p*SK1pK2pK3 + k_u*SK1pK2pK3p
)

SK1pK2pK3p_prime = (
    k_a3*SK1pK2p*K3p - k_d*SK1pK2pK3p
    + k_p*SK1pK2pK3 - k_u*SK1pK2pK3p
)
###
```

We will again examine the same steady state responses as done previously for our new model. The next section of code again varies the concentration of the scaffold protein, S, and reports the output signal $K_3p/K_{3\,total}$.

9. (3 points) Run the code in the section "STEADY-STATE responses" below the ordered model and paste your graph.



10. (5 points) How do the results differ from the unordered binding case? Why might this be?

With the unordered model, there is a rise to a peak, and then it declines as scaffold concentration increases. The ordered model in comparison rises and then approaches a plateau. Ordered binding keeps K2 and K3 from binding to scaffolds that don't have the required upstream kinases, which reduces their separation across different scaffolds. It also preserves the assembled complexes and protects the bound K2p from dephosphorylation while K3 is present. These rules help maintain productive signaling and it prevents the prozone effect seen in the unordered model.

Part 4: Multisite phosphorylation

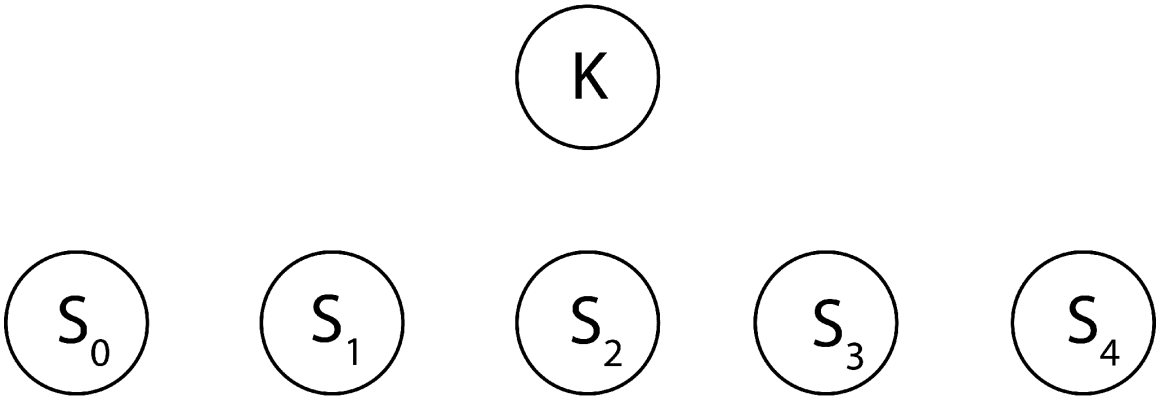
As we discussed in class, many proteins have multiple phosphorylation sites. We will make a simple model of multisite phosphorylation in order to explore the behavior of these systems.

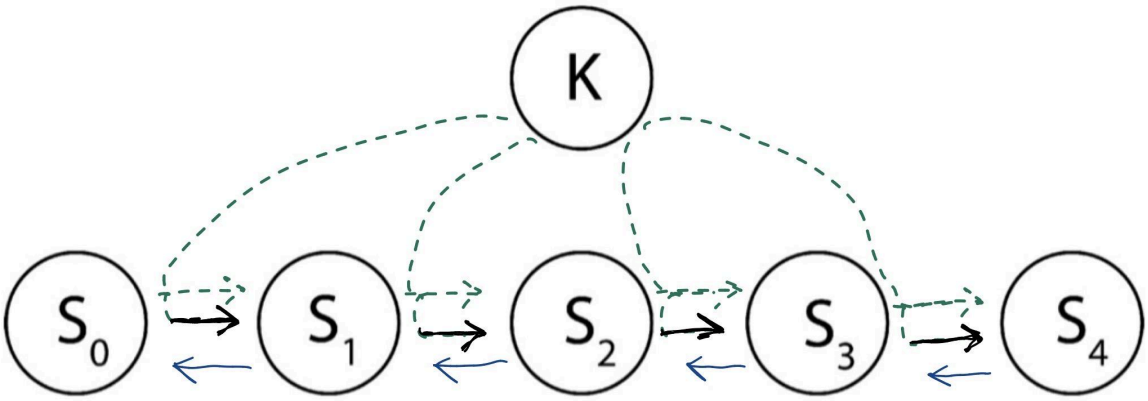
Consider a substrate protein S with 4 phosphorylation sites. As described in class, in this model we won't worry too much about *which* sites are phosphorylated, just about *how many* positions in the protein are phosphorylated. We will use the state variable S_0 to denote the concentration of the substrate with no sites phosphorylated, S_1 for the case with one site phosphorylated, S_2 for two sites phosphorylated, etc.

We will make the following assumptions for this model: (1) the kinase *cannot be saturated* and (2) we won't model the phosphatase explicitly, but will model dephosphorylation as a spontaneous first-order reaction. We will call the kinase K in this model. Note that these assumptions are *exactly* the same as the ones we used for the kinase cascades we modeled in the previous homework.

Answer the questions below:

11. (3 points) Fill in the arrows below to complete the network diagram for this system, using the rules we have used so far in class. Specifically, a dashed line is used to denote catalysis, while solid lines are used to denote actual transformations. Also remember that, if we have a simple first-order transformation that doesn't require a catalyst, we just draw an arrow between the relevant species.





12. (3 points) Now, write out the *chemical reactions* for this model. Remember, these are chemical reactions, **not** ODEs/change equations. Make sure to number your reactions. Hint: you should have 8 equations!

Phosphorylation:

1) $S_0 + K \xrightarrow{k_a} S_1 + K$

2) $S_1 + K \xrightarrow{k_a} S_2 + K$

3) $S_2 + K \xrightarrow{k_a} S_3 + K$

4) $S_3 + K \xrightarrow{k_a} S_4 + K$

Dephosphorylation:

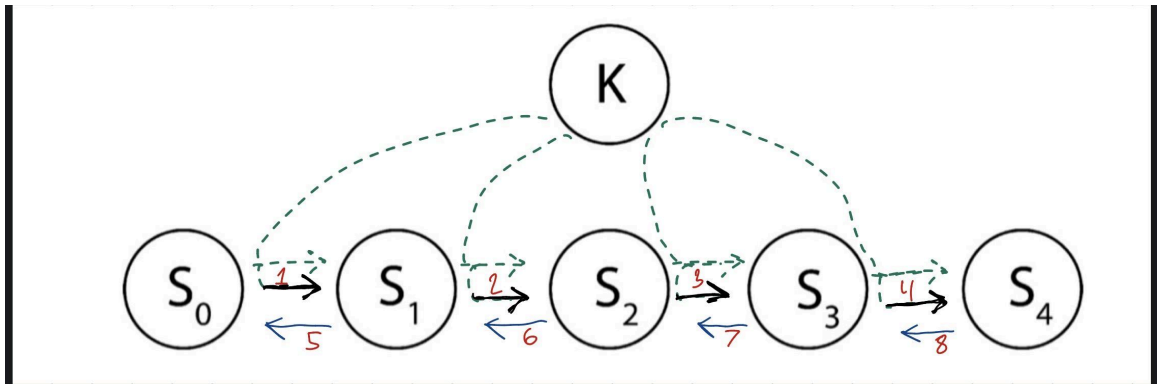
5) $S_1 \xrightarrow{k_u} S_0$

6) $S_2 \xrightarrow{k_u} S_1$

7) $S_3 \xrightarrow{k_u} S_2$

8) $S_4 \xrightarrow{k_u} S_3$

13. (3 points) Copy your drawing from question 11 below. Now, add the number for each of your chemical reactions in your answer to question 12 next to the corresponding arrow in the network diagram.



14. (5 points) In the template Jupyter notebook, you will find a section labeled "Multisite phosphorylation: 4-site model." Write down the change equations for S0, S1, S2, etc. and paste a screenshot of your finalized model below:

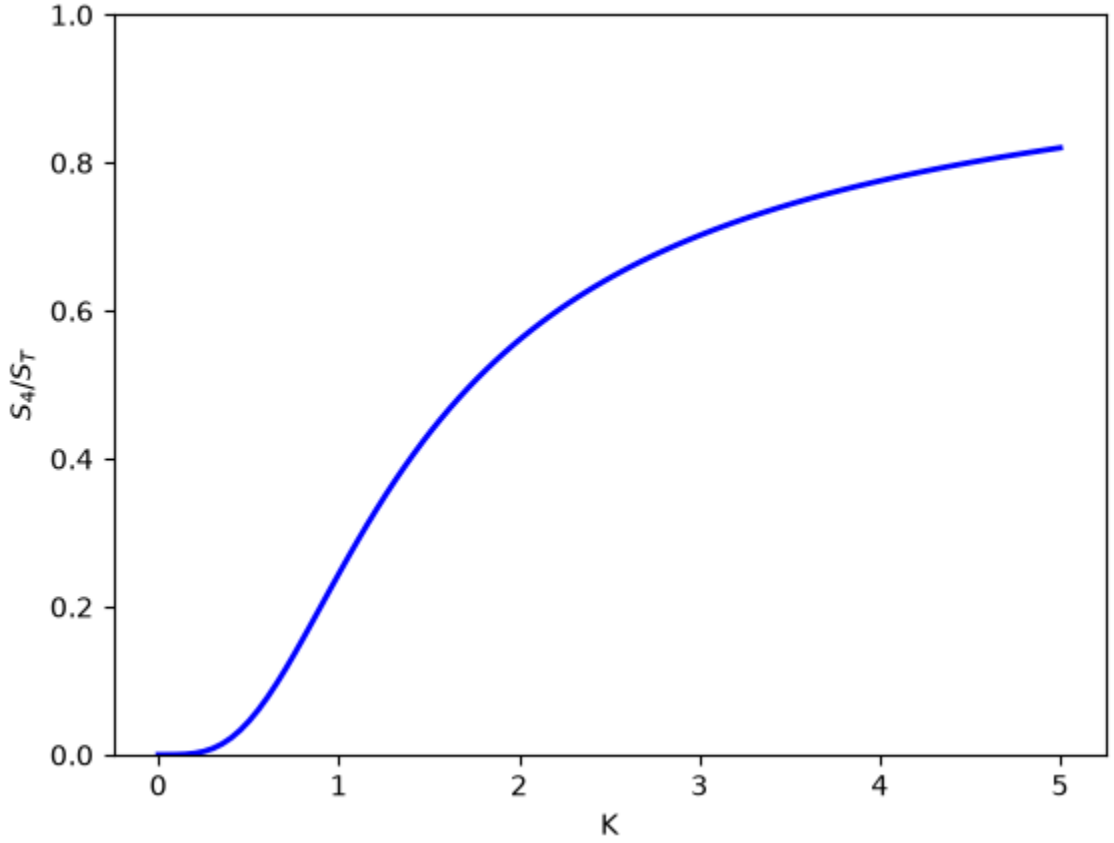
```
#This defines our multt site phosphorylation model. Please run this
def multisite4(states,t,k_a,k_u,K1star):
    S0 = states[0]
    S1 = states[1]
    S2 = states[2]
    S3 = states[3]
    S4 = states[4]
    #The above just makes it easier to relate our ODEs to the ones i
    #creating your ODEs
    # k_a is you assocation constation while k_u is your dissoctiati
    # Fill in your ODE's below


    S0_prime = -k_a*K1star*S0 + k_u*S1
    S1_prime = (
        k_a*K1star*S0 - k_a*K1star*S1 - k_u*S1 + k_u*S2
    )
    S2_prime = (
        k_a*K1star*S1 - k_a*K1star*S2 - k_u*S2 + k_u*S3
    )
    S3_prime = (
        k_a*K1star*S2 - k_a*K1star*S3 - k_u*S3 + k_u*S4
    )
    S4_prime = k_a*K1star*S3 - k_u*S4


    #
    states_prime = [S0_prime,S1_prime,S2_prime,S3_prime,S4_prime]
    return states_prime
```

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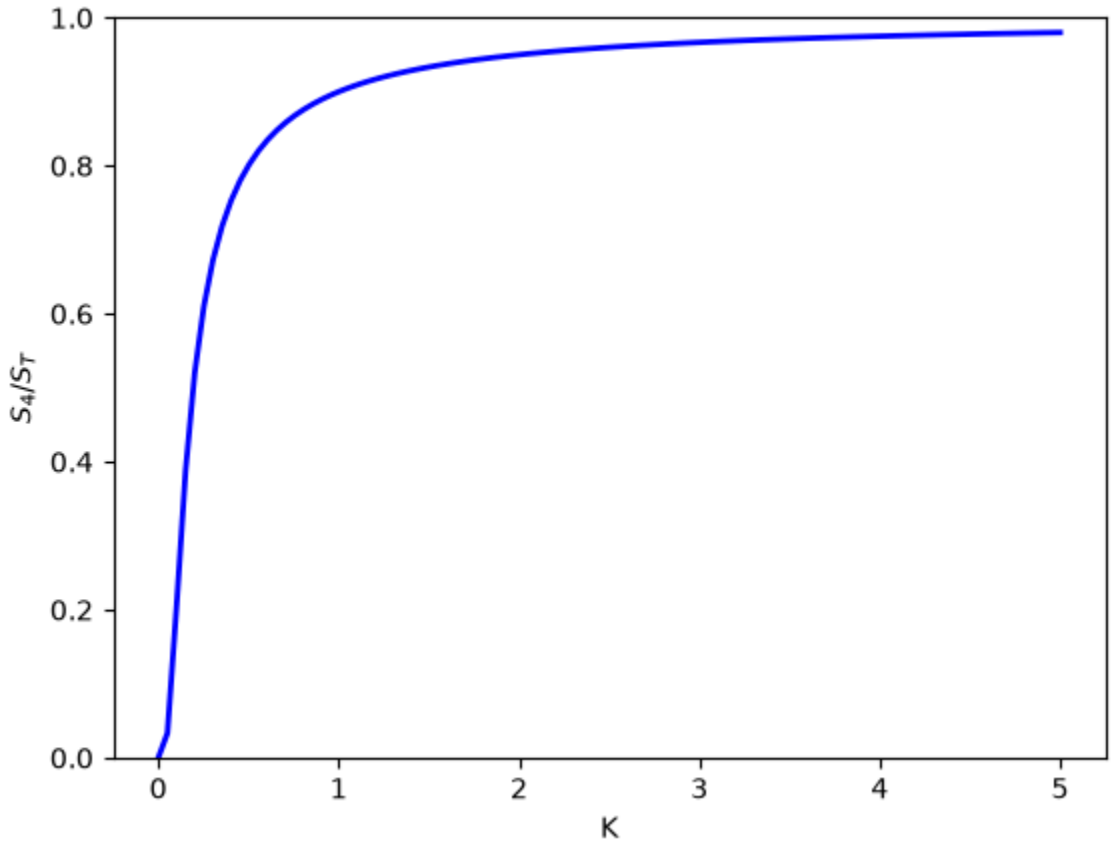
15. (3 points) In the template Jupyter notebook, you will find a section labeled “STEADY-STATE multisite phosphorylation results.” This code changes the concentration of the kinase, , and plots the steady-state fraction of fully-phosphorylated substrate . The default parameters we want to use are already included for you. Run this code and paste the resulting graph below:



16. (5 points) In class, we discussed how multisite phosphorylation can lead to a "threshold." Does your result above look like it has a threshold? Explain your answer.

The graph does show a gradual threshold response. At low K, little substrate reaches S4 since four phosphorylation steps are needed and dephosphorylation is competing with each step. As K increases, phosphorylation becomes faster which allows for more fully phosphorylated substrates to accumulate. The response hence has a low output region, followed by a strong rise, although the transition is smooth rather than an abrupt switch like behavior.

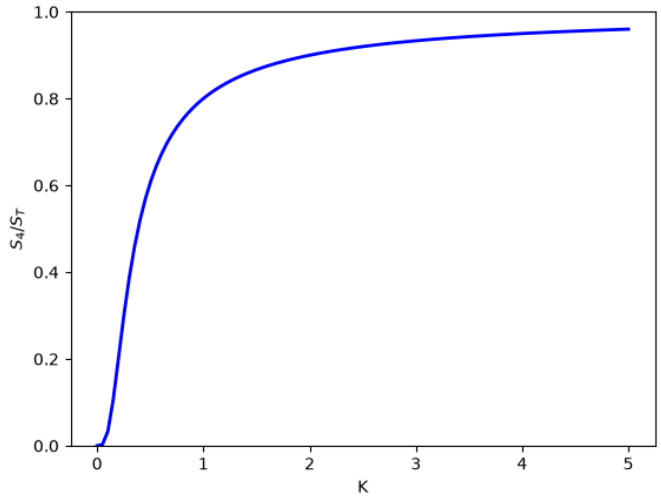
17. (3 points) The default parameters we used for the code above had the activation rate set at and . In other words, the activation rate constant is just a little bit larger than the de-activation rate. Try setting and re-run the code. Paste your result below:



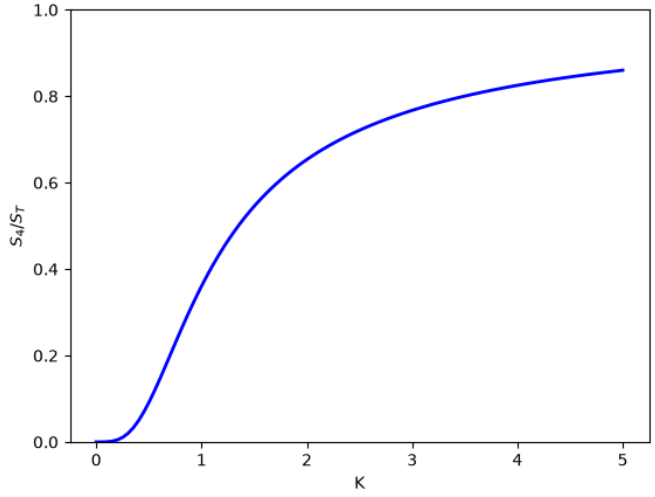
18. (5 points) Compare your results to the ones you got for the original parameters. Do both cases look like they have a threshold? Try changing to different values, say or . As the parameter increases, do you get more of a threshold-like behavior?

In both cases, there is a gradual threshold. But with ku = 1, it occurs at a relatively low K, and is less visible on the plotted scale. When increasing ku to 2, to 7, and to 9, the low output region is extended, and the response is shifted toward a higher K. Stronger dephosphorylation means that more kinase is needed to accumulate fully phosphorylated substrates. This makes the threshold more noticeable without necessarily creating an abrupt switch.

2:



7:



19. (1 point) Together with the rest of your learning team, come up with a fun name for your learning team. Write that name down in the space below, along with the names and emails of the people in your learning team.

Team name: Kindases
Group: Zain (me,), Lia ( ), and Alexis ( )